

ECE 332: Engineering Electromagnetics II

Lecture 11

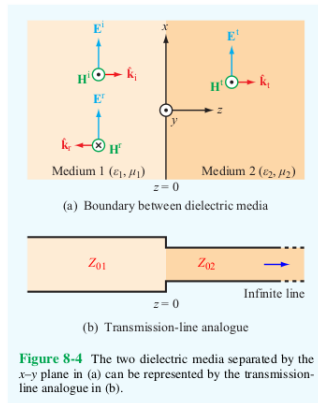
Mark A. Martin

Portland State University

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Last Lecture

- The case of two transmission lines – a lossless line (Line 1) and an infinitely-long lossless line (Line 2) that meet at $z = 0$, is analogous to the case of a normally incident EM wave at a boundary between materials with different EM properties.
- The transmission-line quantities $(\tilde{V}, \tilde{I}, \beta, Z_0)$ and plane-wave quantities $(\tilde{E}, \tilde{H}, k, \eta)$ **satisfy the same equations**, respectively.



Last Lecture (Continued)

- Lessons learned for transmission lines in Chapter 2 can be directly applied to normally-incident EM waves.

Table 8-1 Analogy between plane-wave equations for normal incidence and transmission-line equations, both under lossless conditions.

Plane Wave [Fig. 8-4(a)]	Transmission Line [Fig. 8-4(b)]
$\tilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^i (e^{-jk_1 z} + \Gamma e^{jk_1 z})$ (8.11a)	$\tilde{V}_1(z) = V_0^+ (e^{-j\beta_1 z} + \Gamma e^{j\beta_1 z})$ (8.11b)
$\tilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E_0^i}{\eta_1} (e^{-jk_1 z} - \Gamma e^{jk_1 z})$ (8.12a)	$\tilde{I}_1(z) = \frac{V_0^+}{Z_{01}} (e^{-j\beta_1 z} - \Gamma e^{j\beta_1 z})$ (8.12b)
$\tilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^i e^{-jk_2 z}$ (8.13a)	$\tilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2 z}$ (8.13b)
$\tilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \frac{E_0^i}{\eta_2} e^{-jk_2 z}$ (8.14a)	$\tilde{I}_2(z) = \tau \frac{V_0^+}{Z_{02}} e^{-j\beta_2 z}$ (8.14b)
$\Gamma = (\eta_2 - \eta_1) / (\eta_2 + \eta_1)$	$\Gamma = (Z_{02} - Z_{01}) / (Z_{02} + Z_{01})$
$\tau = 1 + \Gamma$	$\tau = 1 + \Gamma$
$k_1 = \omega \sqrt{\mu_1 \epsilon_1}, \quad k_2 = \omega \sqrt{\mu_2 \epsilon_2}$	$\beta_1 = \omega \sqrt{\mu_1 \epsilon_1}, \quad \beta_2 = \omega \sqrt{\mu_2 \epsilon_2}$
$\eta_1 = \sqrt{\mu_1 / \epsilon_1}, \quad \eta_2 = \sqrt{\mu_2 / \epsilon_2}$	Z_{01} and Z_{02} depend on transmission-line parameters

Last Lecture (Continued)

- If the characteristic impedances of Line 1 and Line 2 are Z_{01} and Z_{02} , resp., the **voltage reflection coefficient** at $z = 0$ from Line 1 to Line 2

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}} \quad (1)$$

is analogous to the **reflection coefficient** Γ for the EM wave.

- The incident and reflected waves create a **standing wave pattern** in Medium 1, described by the **standing-wave ratio**

$$S = \frac{|\tilde{E}_1|_{\max}}{|\tilde{E}_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \quad (2)$$

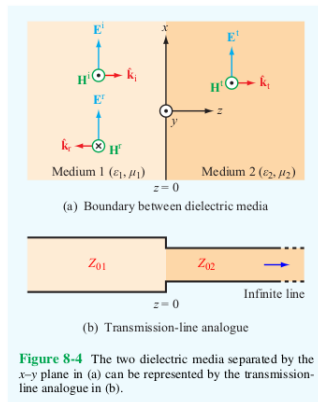


Figure 8-4 The two dielectric media separated by the x - y plane in (a) can be represented by the transmission-line analogue in (b).

Last Lecture (Continued)

- The **locations** $\ell_{\max}$ where the **electric field intensity reaches its maximum value** in Medium 1 are

$$\ell_{\max} = -z = \frac{\theta_{\Gamma} + 2n\pi}{2k_1} = \frac{\theta_{\Gamma}\lambda_1}{4\pi} + \frac{n\lambda_1}{2}, \quad (3)$$

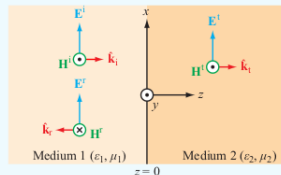
where

$$\Gamma = |\Gamma|e^{j\theta_{\Gamma}} \text{ with } \theta_{\Gamma} \in (-\pi, \pi], \quad (4)$$

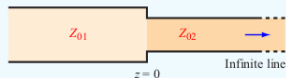
$$\lambda_1 = 2\pi/k_1, \quad (5)$$

and

$$n = \begin{cases} 1, 2, \dots & \theta_{\Gamma} < 0 \\ 0, 1, 2, \dots & \theta_{\Gamma} \geq 0. \end{cases} \quad (6)$$



(a) Boundary between dielectric media



(b) Transmission-line analogue

Figure 8-4 The two dielectric media separated by the x - y plane in (a) can be represented by the transmission-line analogue in (b).

Last Lecture (Continued)

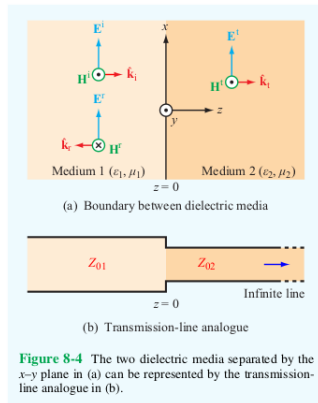
- The **locations** $\ell_{\min}$ where the **electric field intensity reaches its minimum value** in Medium 1 are

$$\ell_{\min} = \begin{cases} \ell_{\max} + \lambda_1/4, & \ell_{\max} < \lambda_1/4 \\ \ell_{\max} - \lambda_1/4, & \ell_{\max} \geq \lambda_1/4. \end{cases} \quad (7)$$

- When the media are **lossless**,

$$\theta_{\Gamma} = \begin{cases} 0, & \eta_2 > \eta_1 \\ \pi, & \eta_2 < \eta_1. \end{cases} \quad (8)$$

- The expressions for $\ell_{\max}$ and $\ell_{\min}$ are also valid when Medium 1 is a **low-loss dielectric** and Medium 2 is a **dielectric or conductor**.



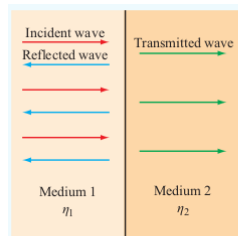
Last Lecture (Continued)

Power Flow in Lossless Media

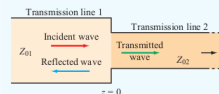
We continued discussing what happens when an **x-polarized EM wave** traveling in the $+z$ direction **normal** to the **boundary between two lossless media** intersects the boundary, which is located at the x - y plane, and is **partially transmitted** and **partially reflected**.

Medium 1

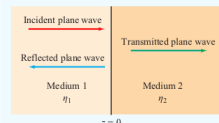
- Fills half-space $z \leq 0$
- Permittivity ϵ_1
- Permeability μ_1
- $k_1 = \omega\sqrt{\mu_1\epsilon_1}$, $\eta_1 = \sqrt{\mu_1/\epsilon_1}$



(a) Normal incidence



(a) Boundary between transmission lines



(b) Boundary between different media

Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

Last Lecture (Continued)

Medium 2

- Fills half-space $z \geq 0$
- Permittivity ϵ_2
- Permeability μ_2
- $k_2 = \omega\sqrt{\mu_2\epsilon_2}$, $\eta_2 = \sqrt{\mu_2/\epsilon_2}$

Incident Wave ($\mathbf{E}^i, \mathbf{H}^i$)

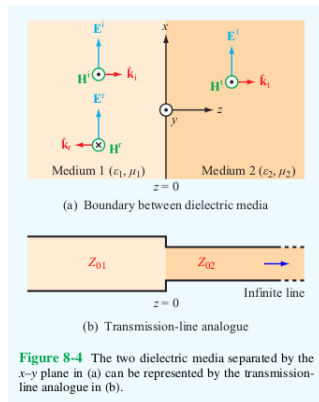
$$\tilde{\mathbf{E}}^i(z) = E_0^i e^{-jk_1 z} \hat{\mathbf{x}} \quad (9)$$

$$\tilde{\mathbf{H}}^i(z) = (E_0^i/\eta_1) e^{-jk_1 z} \hat{\mathbf{y}} \quad (10)$$

Reflected Wave ($\mathbf{E}^r, \mathbf{H}^r$)

$$\tilde{\mathbf{E}}^r(z) = \Gamma E_0^i e^{jk_1 z} \hat{\mathbf{x}} \quad (11)$$

$$\tilde{\mathbf{H}}^r(z) = -(\Gamma E_0^i/\eta_1) e^{jk_1 z} \hat{\mathbf{y}} \quad (12)$$



Last Lecture (Continued)

Transmitted Wave ($\mathbf{E}^t, \mathbf{H}^t$)

$$\tilde{\mathbf{E}}^t(z) = \tau E_0^i e^{-jk_2 z} \hat{\mathbf{x}} \quad (13)$$

$$\tilde{\mathbf{H}}^t(z) = (\tau E_0^i / \eta_2) e^{-jk_2 z} \hat{\mathbf{y}} \quad (14)$$

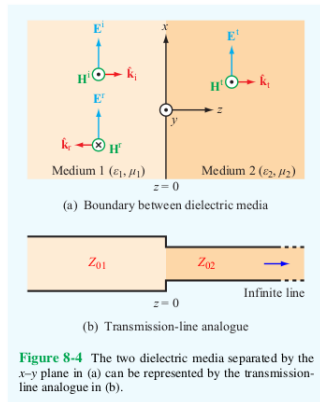
Here, the **reflection coefficient** is

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (15)$$

and the **transmission coefficient** is

$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1} = 1 + \Gamma. \quad (16)$$

When the media are **lossless**, Γ and τ are **real-valued**.



Last Lecture (Continued)

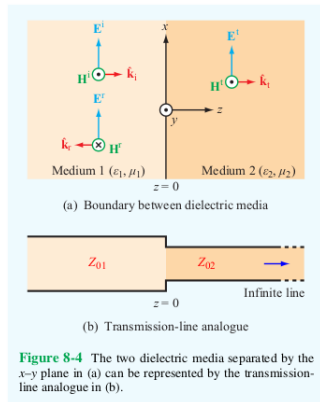
The **electric** and **magnetic fields** in **Medium 1** are

$$\tilde{\mathbf{E}}_1(z) = E_0^i \left(e^{-jk_1 z} + \Gamma e^{jk_1 z} \right) \hat{\mathbf{x}} \quad (17)$$

$$\tilde{\mathbf{H}}_1(z) = \frac{E_0^i}{\eta_1} \left(e^{-jk_1 z} - \Gamma e^{jk_1 z} \right) \hat{\mathbf{y}} \quad (18)$$

and the **average power density** in **Medium 1** is

$$\begin{aligned} \mathbf{S}_{\text{av}1} &= \mathbf{S}_{\text{av}}^i + \mathbf{S}_{\text{av}}^r \\ &= \frac{|E_0^i|^2}{2\eta_1} \left(1 - |\Gamma|^2 \right) \hat{\mathbf{z}}. \end{aligned} \quad (19)$$



Last Lecture (Continued)

The **average power density** in **Medium 2** is

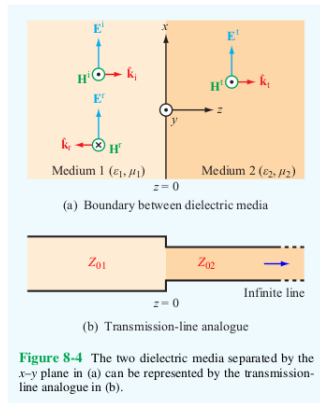
$$\mathbf{S}_{av2} = \frac{|\tau|^2 |E_0^i|^2}{2\eta_2} \hat{\mathbf{z}}. \quad (20)$$

Since (τ and Γ are real-valued and)

$$\frac{\tau^2}{\eta_2} = \frac{1 - \Gamma^2}{\eta_1}, \quad (21)$$

$$\mathbf{S}_{av2} = \frac{\tau^2}{\eta_2} \frac{|E_0^i|^2}{2} \hat{\mathbf{z}} = \left(\frac{1 - \Gamma^2}{\eta_1} \right) \frac{|E_0^i|^2}{2} \hat{\mathbf{z}} = \mathbf{S}_{av1}, \quad (22)$$

as expected from **power conservation**.



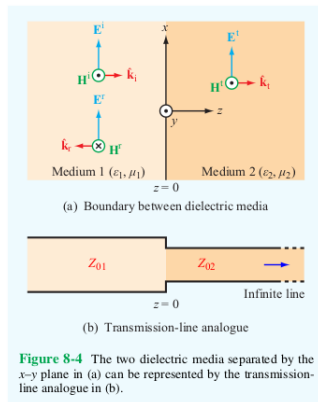
Last Lecture (Continued)

- We explored an example of yellow light normally incident on glass.

Some possible applications of this case are

- windows
- lenses

Note that η_1 , η_2 , Γ , and τ are **independent of frequency** for **lossless media**. However, k_1 and k_2 are **frequency dependent** and $\mathbf{S}_{av1}$ and $\mathbf{S}_{av2}$ only depend on frequency through $|E_0^i|$.



Last Lecture (Continued)

Normal Incidence in Lossy Media

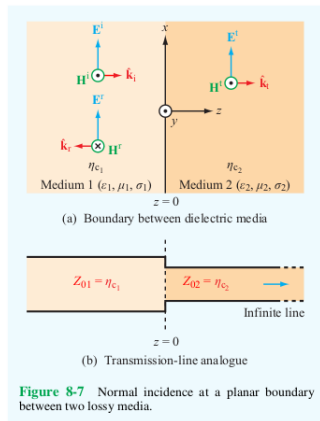
In a **lossy medium**, the **intrinsic impedance** η_c and **propagation constant** $\gamma = \alpha + j\beta$ are **complex-valued**.

Medium 1: $\epsilon_1, \mu_1, \sigma_1$,

$$\eta_{c1} = \epsilon_1 - j\sigma_1/\omega, \quad \gamma_1 = \alpha_1 + j\beta_1$$

$$\tilde{\mathbf{E}}_1(z) = E_0^i (e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}) \hat{\mathbf{x}} \quad (23)$$

$$\tilde{\mathbf{H}}_1(z) = \frac{E_0^i}{\eta_{c1}} (e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}) \hat{\mathbf{y}} \quad (24)$$



Last Lecture (Continued)

Medium 2: $\epsilon_2, \mu_2, \sigma_2,$

$$\eta_{c2} = \epsilon_2 - j\sigma_2/\omega, \quad \gamma_2 = \alpha_2 + j\beta_2$$

$$\tilde{\mathbf{E}}_2(z) = \tau E_0^i e^{-\gamma_2 z} \hat{\mathbf{x}} \quad (25)$$

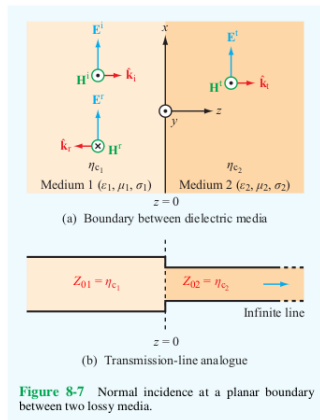
$$\tilde{\mathbf{H}}_2(z) = \frac{\tau E_0^i}{\eta_{c2}} e^{-\gamma_2 z} \hat{\mathbf{y}} \quad (26)$$

As before, the **reflection and transmission coefficients** are

$$\Gamma = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}} \quad (27)$$

and

$$\tau = 1 + \Gamma = \frac{2\eta_{c2}}{\eta_{c2} + \eta_{c1}}. \quad (28)$$



Last Lecture (Continued)

- We considered an example of radio waves normally incident on a metal surface.

Some possible applications of this case are

- mirrors
- shielding
- filters

Note that η_{c1} , η_{c2} , Γ , τ , γ_1 , and γ_2 are **frequency dependent** for **lossy media**.

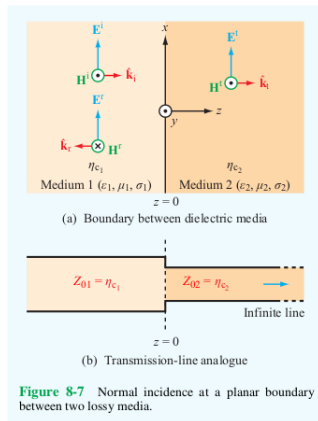


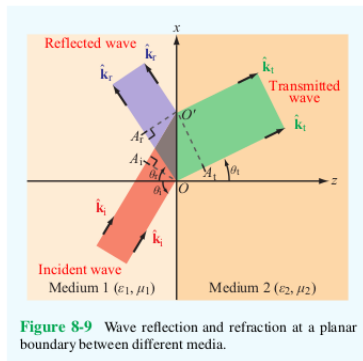
Figure 8-7 Normal incidence at a planar boundary between two lossy media.

Last Lecture (Continued)

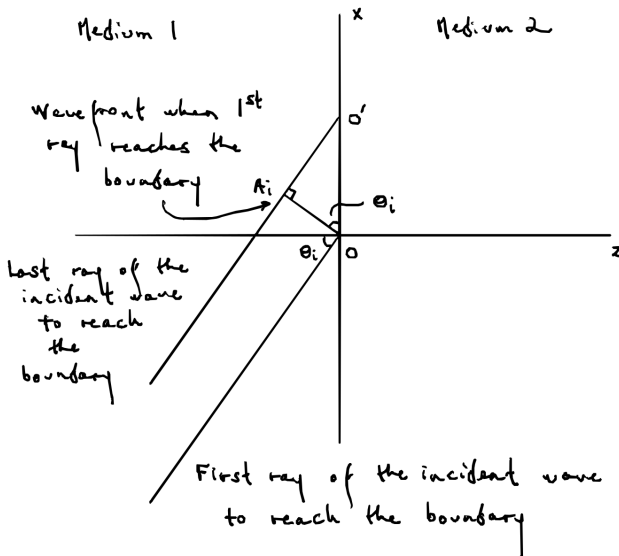
Snell's Laws

For **lossless** media, when the incident wave is traveling in a direction that isn't normal to the boundary, the rays of the waves **travel in the directions below** and **intersect the boundary at the given angles relative to the normal to the boundary**.

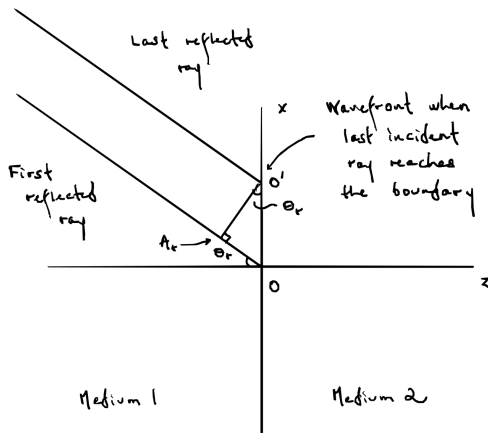
- **incident wave:**
direction $\hat{\mathbf{k}}_i$, angle $\theta_i \in [0, \pi/2]$,
- **reflected wave:**
direction $\hat{\mathbf{k}}_r$, angle $\theta_r \in [0, \pi/2]$,
- **transmitted wave:**
direction $\hat{\mathbf{k}}_t$, angle $\theta_t \in [0, \pi/2]$.



Last Lecture (Continued)

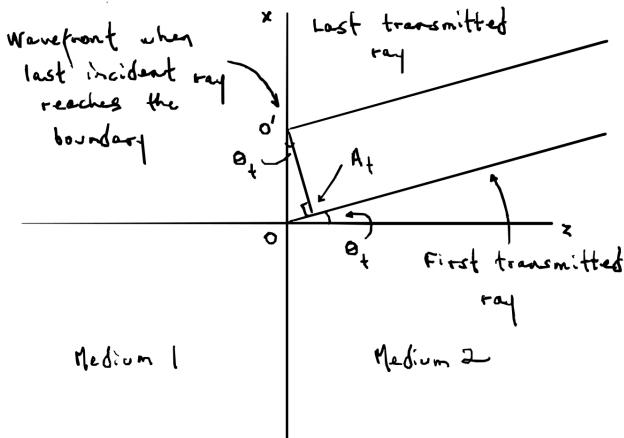


Last Lecture (Continued)



The time it takes for the last ray of the incident wave to reach the boundary to go from point A_i to point O' is the same as the time the first reflected ray travels from point O to point A_r .

Last Lecture (Continued)



The time it takes for the last ray of the incident wave to reach the boundary to go from point A_i to point O' is the same as the time the first transmitted ray travels from point O to point A_t .

Last Lecture (Continued)

- Under these circumstances,

$$\theta_r = \theta_i \quad (\text{Snell's law of } \mathbf{reflection}) \quad (29)$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \sqrt{\frac{\mu_{r1} \epsilon_{r1}}{\mu_{r2} \epsilon_{r2}}} \quad (\text{Snell's law of } \mathbf{refraction}). \quad (30)$$

Snell's Laws (Continued)

The **index of refraction** of a medium with phase velocity u_p is

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} = \sqrt{\mu_r\epsilon_r}. \quad (31)$$

Therefore, altogether

$$\theta_r = \theta_i \quad (\text{Snell's law of } \mathbf{reflection}) \quad (32)$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1\epsilon_1}{\mu_2\epsilon_2}} = \sqrt{\frac{\mu_{r1}\epsilon_{r1}}{\mu_{r2}\epsilon_{r2}}} \quad (\text{Snell's law of } \mathbf{refraction}). \quad (33)$$

For **nonmagnetic materials**, $\mu_1 = \mu_2 = \mu_0$ and the **law of refraction** is

$$\frac{\sin \theta_t}{\sin \theta_i} = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \frac{\eta_2}{\eta_1}. \quad (34)$$

Snell's Laws (Continued)

Denser nonmagnetic materials often have higher permittivities and, hence, higher indices of refraction. So, nonmagnetic materials with **higher indices of refraction** are often called **denser** compared to nonmagnetic materials with **lower indices of refraction**.

As a sanity check, suppose $\theta_i = 0$. This is the case of **normal incidence**. In this case,

$$\theta_r = \theta_i = 0 \quad (35)$$

and

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i = 0 \implies \theta_t = 0, \quad (36)$$

regardless of the relationship between n_1 and n_2 , as expected.

More generally, when $0 < \theta_i \leq \pi/2$ and $n_2 > n_1 > 0$,

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i < \sin \theta_i \implies \theta_t < \theta_i \quad (37)$$

Snell's Laws (Continued)

whereas, when $0 < \theta_i \leq \pi/2$ and $n_1 > n_2 > 0$,

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i > \sin \theta_i \implies \theta_t > \theta_i \quad (38)$$

So, when $\theta_i \neq 0$ and $n_1 < n_2$, the transmitted wave is directed toward the normal to the boundary, which is called **inward refraction**, and, when $\theta_i \neq 0$ and $n_1 > n_2$, the transmitted wave is directed away from the normal to the boundary, which is called **outward refraction**.

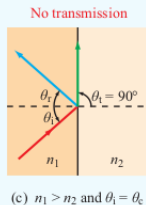
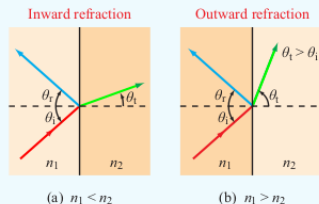


Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin \theta_t = (n_1/n_2) \sin \theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; and (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

Snell's Laws (Continued)

When $n_1 > n_2$, there will be an angle θ_i where $\theta_t = \pi/2$. This is called the **critical angle** θ_c and it satisfies

$$\sin \theta_c = \frac{n_2}{n_1} \sin \left(\frac{\pi}{2} \right) = \frac{n_2}{n_1}. \quad (39)$$

If $\mu_1 = \mu_2$,

$$\sin \theta_c = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}}. \quad (40)$$

When $n_1 > n_2$ and $\theta_i \geq \theta_c$, the incident wave is entirely reflected, which is called **total internal reflection**.

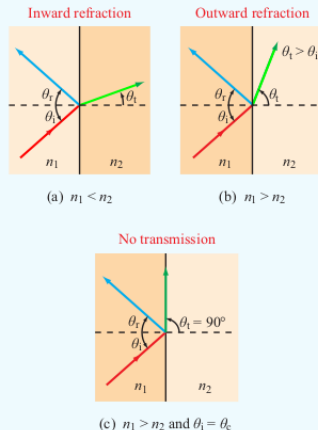


Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin \theta_t = (n_1/n_2) \sin \theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; and (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

Fiber Optics

Small-diameter cylinders of dielectric material, typically glass or plastic, can efficiently carry light via the phenomenon of total internal reflection.

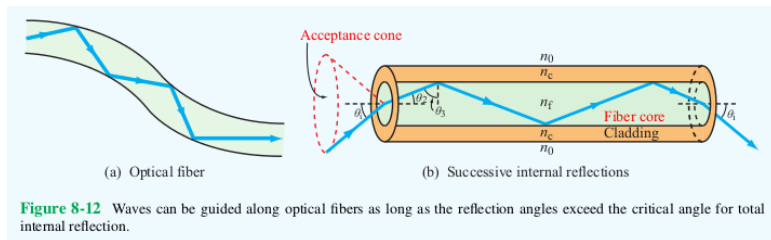


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Here, the **fiber core** has an index of refraction n_f that is higher than the index of refraction n_c of the surrounding **cladding**. The critical angle for the fiber is given by

$$\sin \theta_c = \frac{n_c}{n_f}. \quad (41)$$

A light ray traveling through the fiber will intersect the outer surface of the cylinder at an angle $\theta_3 \in (0, \pi/2]$ relative to the normal to the surface.

Fiber Optics (Continued)

To ensure total internal reflection, θ_3 must satisfy

$$\sin \theta_3 \geq \sin \theta_c = \frac{n_c}{n_f}. \quad (42)$$

The angle that the ray makes with the central axis of the fiber is

$$\theta_2 = \frac{\pi}{2} - \theta_3. \quad (43)$$

So, for the light to be totally internally reflected, θ_2 must satisfy

$$\cos \theta_2 = \cos \left(\frac{\pi}{2} - \theta_3 \right) = \sin \theta_3 \geq \frac{n_c}{n_f}. \quad (44)$$

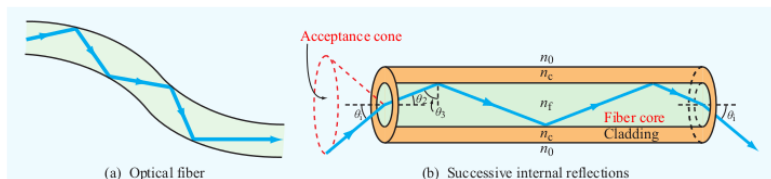


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Fiber Optics (Continued)

If the index of refraction of the medium outside of the fiber is n_0 and a ray of light shining onto the end of the fiber intersects the fiber core at an angle θ_i relative to the normal to the end (i.e. the direction of the central axis of the core), then, according to the law of refraction,

$$\sin \theta_2 = \frac{n_0}{n_f} \sin \theta_i. \quad (45)$$

By a trigonometric identity,

$$1 = \cos^2 \theta_2 + \sin^2 \theta_2 = \cos^2 \theta_2 + \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_i. \quad (46)$$

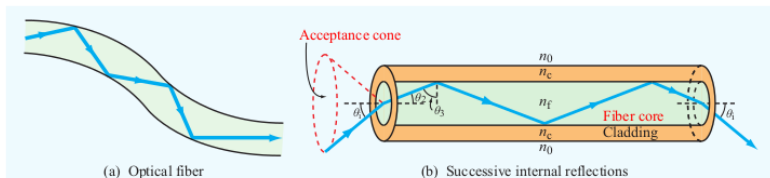


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

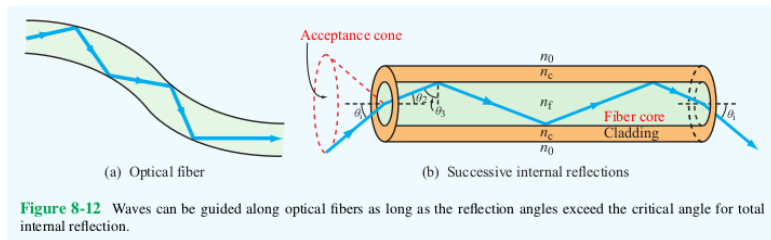
Fiber Optics (Continued)

Solving for $\cos \theta_2$,

$$\cos \theta_2 = \left(1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_i \right)^{1/2}. \quad (47)$$

So, for total internal reflection, θ_i must satisfy

$$\left(1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_i \right)^{1/2} \geq \frac{n_c}{n_f}. \quad (48)$$



Fiber Optics (Continued)

Squaring both sides of this inequality,

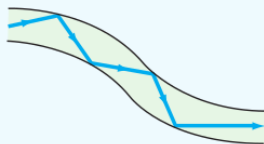
$$1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2 \theta_i \geq \frac{n_c^2}{n_f^2} \implies \left(\frac{n_0}{n_f}\right)^2 \sin^2 \theta_i \leq 1 - \frac{n_c^2}{n_f^2} \quad (49)$$

so that

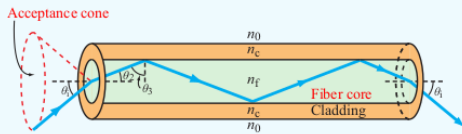
$$\sin^2 \theta_i \leq \left(\frac{n_f}{n_0}\right)^2 \left(1 - \frac{n_c^2}{n_f^2}\right) = \frac{1}{n_0^2} (n_f^2 - n_c^2) \quad (50)$$

and, finally, taking the square root of both sides

$$\sin \theta_i \leq \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2}. \quad (51)$$



(a) Optical fiber



(b) Successive internal reflections

Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

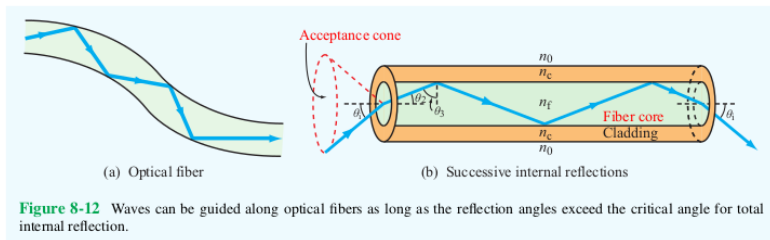
Fiber Optics (Continued)

Maximum Acceptance Cone

The **largest angle** θ_i that produces total internal reflection in the fiber is the **acceptance angle** θ_a , which satisfies

$$\sin \theta_a = \frac{1}{n_0} \left(n_f^2 - n_c^2 \right)^{1/2}. \quad (52)$$

The cone containing all rays that intersect the fiber core with $\theta_i \leq \theta_a$ is called the **acceptance cone**.



Fiber Optics (Continued)

Rays entering the fiber core at different angles θ_i follow different paths through the fiber. Each path is called a **mode** and different modes travel through the fiber in different amounts of time. This is called **modal dispersion** and distorts pulses sent through the fiber. This causes rectangular pulses of initial width τ_i and period T to be spread out into rounded pulses of width τ . If $\tau > T$, the pulses arriving at the end of the fiber will overlap and it might not be possible to interpret the signal. To prevent this problem, the pulses are usually created with a period $T \geq 2\tau$.

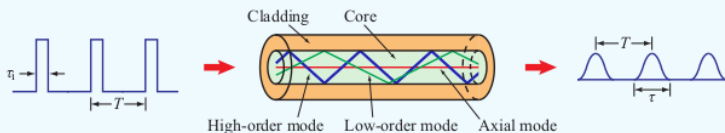


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

Fiber Optics (Continued)

Here,

$$\cos \theta_2 = n_c / n_f \quad (53)$$

and the maximum distance a ray will travel through a fiber of length ℓ is

$$\ell_{\max} = \frac{\ell}{\cos \theta_2} = \ell n_f / n_c \quad (54)$$

So, the travel time for the longest path will be

$$t_{\max} = \ell_{\max} / u_p = \frac{\ell n_f^2}{c n_c}. \quad (55)$$

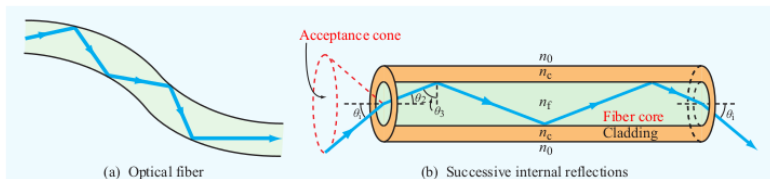


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Fiber Optics (Continued)

The minimum travel time will be

$$t_{\min} = \frac{\ell}{u_p} = \frac{\ell n_f}{c} \quad (56)$$

and the maximum time delay will be

$$\tau = \Delta t = t_{\max} - t_{\min} = \frac{\ell n_f}{c} \left(\frac{n_f - 1}{n_c} \right). \quad (57)$$

Requiring $T \geq 2\tau$, the **maximum data transmission rate** will then be

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2\ell n_f(n_f - n_c)}. \quad (58)$$

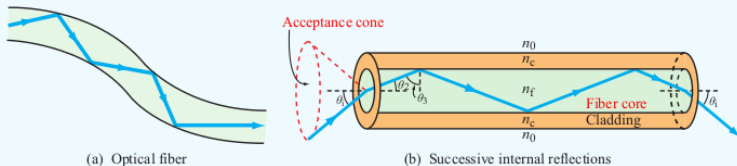


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Fiber Optics (Continued)

The cone consisting of all of the paths that light at the end of the fiber is allowed to follow is called the **entrance cone**.

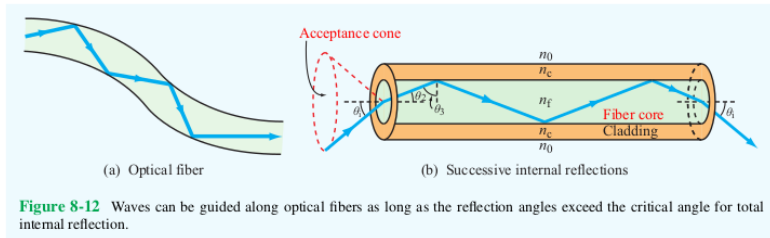


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

If the entrance cone can be restricted to be narrower than the acceptance cone, with maximum angle $\theta_i(\max)$, then the **maximum data transmission rate** can be **increased** to

$$f_p = \frac{c}{2ln_f} \left(\frac{\cos \theta_2(\max)}{1 - \cos \theta_2(\max)} \right), \quad (59)$$

Fiber Optics (Continued)

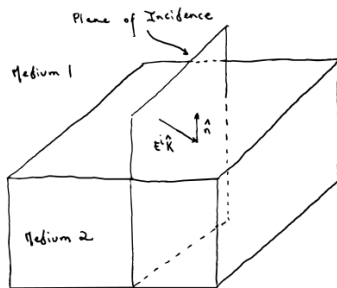
where

$$\cos \theta_2(\max) = \left(1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_i(\max) \right)^{1/2}. \quad (60)$$

Reflection and Transmission at Oblique Incidence

Snell's laws describe what happens when an EM wave **encounters a boundary** between **two lossless media** with different electromagnetic properties **obliquely**, based on a **geometric optics model**.

However, these laws only tell part of the story. A **more complete picture** comes from representing the electric field of the incident wave in terms of components defined relative to the **plane of incidence**. The plane of incidence is the plane formed by the **normal to the boundary $\hat{n}$** and the **direction of propagation of the EM wave $\hat{k}$** .



The media are **both assumed to be lossless**.

Reflection and Transmission at Oblique Incidence (Cont.)

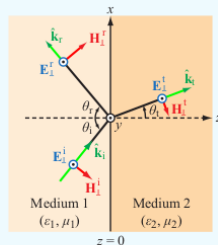
The electric and magnetic fields ($\mathbf{E}$, $\mathbf{H}$) of **any EM wave** (with any polarization) can be written as

$$\mathbf{E} = \mathbf{E}_{\parallel} + \mathbf{E}_{\perp}, \quad \mathbf{H} = \mathbf{H}_{\parallel} + \mathbf{H}_{\perp}, \quad (61)$$

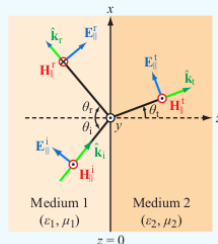
where the polarization of ($\mathbf{E}_{\parallel}$, $\mathbf{H}_{\parallel}$) is **parallel** to the plane of incidence and the polarization of ($\mathbf{E}_{\perp}$, $\mathbf{H}_{\perp}$) is **perpendicular** to the plane of incidence. This means

- $\mathbf{E}_{\parallel}$ is parallel
- and $\mathbf{E}_{\perp}$ is perpendicular

to the plane of incidence.



(a) Perpendicular polarization



(b) Parallel polarization

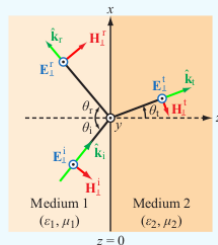
Reflection and Transmission at Oblique Incidence (Cont.)

Since the electric and magnetic fields are orthogonal, this also means

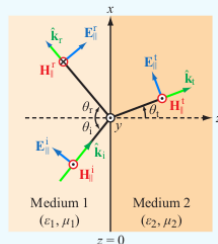
- $\mathbf{H}_{\parallel}$ is perpendicular
- and $\mathbf{H}_{\perp}$ is parallel

to the plane of incidence.

The wave $(\mathbf{E}_{\parallel}, \mathbf{H}_{\parallel})$ is called **transverse magnetic (TM)** since $\mathbf{H}_{\parallel}$ is $\perp$ the plane of incidence and $(\mathbf{E}_{\perp}, \mathbf{H}_{\perp})$ is called **transverse electric (TE)** since $\mathbf{E}_{\perp}$ is $\perp$ the plane of incidence.



(a) Perpendicular polarization



(b) Parallel polarization

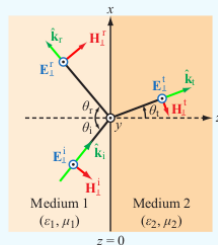
Reflection and Transmission at Oblique Incidence (Cont.)

If the EM fields of the **incident wave** are $(\mathbf{E}^i, \mathbf{H}^i)$, the EM fields of the **reflected wave** are $(\mathbf{E}^r, \mathbf{H}^r)$, and the EM fields of the **transmitted wave** are $(\mathbf{E}^t, \mathbf{H}^t)$, then

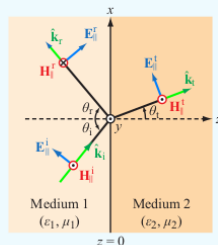
$$\mathbf{E}^i = \mathbf{E}_{\parallel}^i + \mathbf{E}_{\perp}^i, \quad \mathbf{H}^i = \mathbf{H}_{\parallel}^i + \mathbf{H}_{\perp}^i, \quad (62)$$

$$\mathbf{E}^r = \mathbf{E}_{\parallel}^r + \mathbf{E}_{\perp}^r, \quad \mathbf{H}^r = \mathbf{H}_{\parallel}^r + \mathbf{H}_{\perp}^r, \quad (63)$$

$$\mathbf{E}^t = \mathbf{E}_{\parallel}^t + \mathbf{E}_{\perp}^t, \quad \mathbf{H}^t = \mathbf{H}_{\parallel}^t + \mathbf{H}_{\perp}^t. \quad (64)$$



(a) Perpendicular polarization



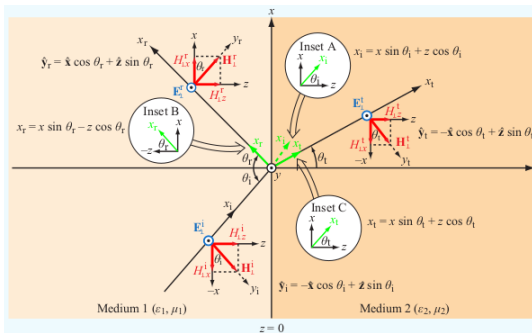
(b) Parallel polarization

Reflection and Transmission at Oblique Incidence (Cont.)

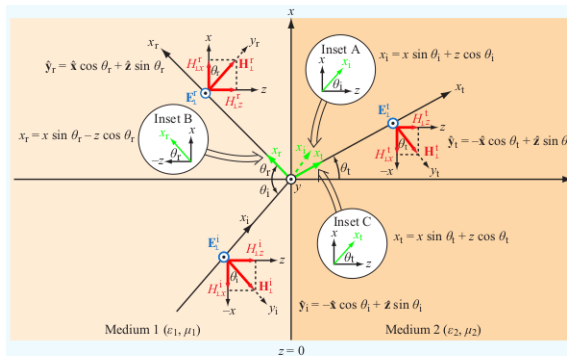
Perpendicular Polarization

Consider a **ray of a perpendicularly polarized incident wave** ($\mathbf{E}_\perp^i, \mathbf{H}_\perp^i$) that intersects a flat boundary between two lossless media and define a **global Cartesian coordinate system** with

- the **origin** at the point where the ray intersects the boundary,
- the **x axis** in the plane of incidence along the boundary,
- the **y axis** be perpendicular to the plane of incidence,
- and the **z axis** in the plane of incidence perpendicular to the boundary with the positive **z** direction pointing into Medium 2.



Reflection and Transmission at Oblique Incidence (Cont.)



Now, define a **local Cartesian coordinate system** associated with the **incident wave** with coordinates (x_i, y_i, y) , where

- $\hat{\mathbf{x}}_i$ is the direction of propagation of the incident wave,
- $\hat{\mathbf{y}}_i$ be the direction of $\tilde{\mathbf{H}}_i^\perp$, which lies in the plane of incidence,
- and $\hat{\mathbf{y}}$ is the same direction as in the global coordinate system.

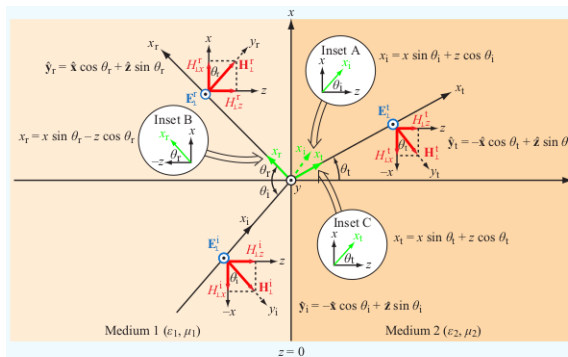
Note that $\tilde{\mathbf{E}}_i^\perp$ is in the $\hat{\mathbf{y}}$ direction, which is $\perp$ the plane of incidence.

Reflection and Transmission at Oblique Incidence (Cont.)

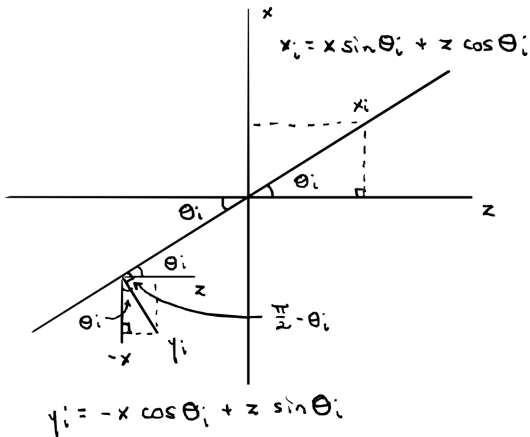
In the local coordinate system,

$$\tilde{\mathbf{E}}_{\perp}^i = E_{\perp 0}^i e^{-jk_1 x_i} \hat{\mathbf{y}}, \quad \tilde{\mathbf{H}}_{\perp}^i = \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1 x_i} \hat{\mathbf{y}}, \quad (65)$$

where $E_{\perp 0}^i = \tilde{\mathbf{E}}_{\perp}^i(x_i = 0)$ and k_1 and η_1 are the wavenumber and intrinsic impedance of Medium 1, respectively.



Reflection and Transmission at Oblique Incidence (Cont.)



Based on the geometry shown above,

$$\mathbf{x}_i = x \sin \theta_i + z \cos \theta_i \quad \text{and} \quad \hat{\mathbf{y}}_i = -\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i. \quad (66)$$

Reflection and Transmission at Oblique Incidence (Cont.)

So, in the **global** x - y - z **coordinate system**, the EM fields of the **incident wave** are

$$\begin{aligned}\tilde{\mathbf{E}}_{\perp}^i &= E_{\perp 0}^i e^{-jk_1 x_i} \hat{\mathbf{y}} \\ &= E_{\perp 0}^i e^{-jk_1 (x \sin \theta_i + z \cos \theta_i)} \hat{\mathbf{y}}\end{aligned}\quad (67)$$

and

$$\begin{aligned}\tilde{\mathbf{H}}_{\perp}^i &= \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1 x_i} \hat{\mathbf{y}}_i \\ &= \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1 (x \sin \theta_i + z \cos \theta_i)} (-\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i).\end{aligned}\quad (68)$$

Reflection and Transmission at Oblique Incidence (Cont.)

Next, define a **local Cartesian coordinate system** associated with the **reflected wave** with coordinates (y_r, x_r, y) , where

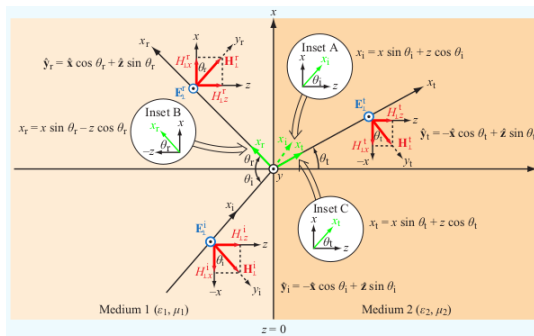
- $\hat{\mathbf{y}}_r$ is the **direction of $\tilde{\mathbf{H}}_r$** , which lies in the plane of incidence,
- $\hat{\mathbf{x}}_r$ is the **direction of propagation** of the **reflected wave**,
- and $\hat{\mathbf{y}}$ is the same direction as in the global coordinate system.

Then $\tilde{\mathbf{E}}_r$ is in the $\hat{\mathbf{y}}$ direction and the EM fields of the **reflected wave** are

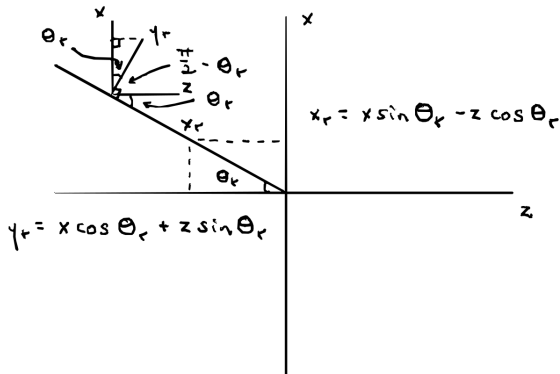
$$\begin{aligned}\tilde{\mathbf{E}}_r &= E_{\perp 0}^r e^{-jk_1 x_r} \hat{\mathbf{y}}, \\ \tilde{\mathbf{H}}_r &= \frac{E_{\perp 0}^r}{\eta_1} e^{-jk_1 x_r} \hat{\mathbf{y}}_r,\end{aligned}\quad (69)$$

where

$E_{\perp 0}^r = \tilde{\mathbf{E}}_r(x_r = 0)$ and k_1 and η_1 are the wavenumber and intrinsic impedance of Medium 1, respectively.



Reflection and Transmission at Oblique Incidence (Cont.)



Based on the geometry shown above,

$$x_r = x \sin \theta_r - z \cos \theta_r \quad \text{and} \quad \hat{\mathbf{y}}_r = \hat{\mathbf{x}} \cos \theta_r + \hat{\mathbf{z}} \sin \theta_r. \quad (70)$$

Reflection and Transmission at Oblique Incidence (Cont.)

So, in the **global** x - y - z **coordinate system**, the EM fields of the **reflected wave** are

$$\begin{aligned}\tilde{\mathbf{E}}_{\perp}^r &= E_{\perp 0}^r e^{-jk_1 x_r} \hat{\mathbf{y}} \\ &= E_{\perp 0}^r e^{-jk_1 (x \sin \theta_r - z \cos \theta_r)} \hat{\mathbf{y}}\end{aligned}\quad (71)$$

and

$$\begin{aligned}\tilde{\mathbf{H}}_{\perp}^r &= \frac{E_{\perp 0}^r}{\eta_1} e^{-jk_1 x_r} \hat{\mathbf{y}}_r \\ &= \frac{E_{\perp 0}^r}{\eta_1} e^{-jk_1 (x \sin \theta_r - z \cos \theta_r)} (\hat{\mathbf{x}} \cos \theta_r + \hat{\mathbf{z}} \sin \theta_r).\end{aligned}\quad (72)$$

Reflection and Transmission at Oblique Incidence (Cont.)

Finally, define a **local Cartesian coordinate system** associated with the **transmitted wave** with coordinates (x_t, y, y_r) , where

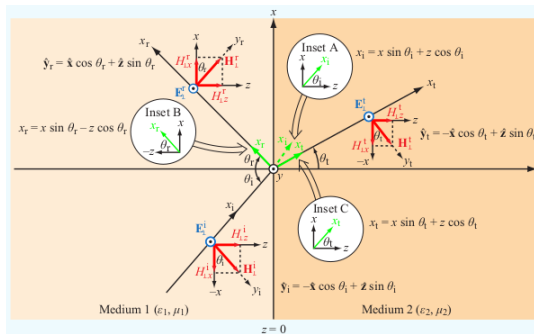
- $\hat{\mathbf{x}}_t$ is the **direction of propagation** of the **transmitted wave**,
- $\hat{\mathbf{y}}$ is the same direction as in the global coordinate system,
- and $\hat{\mathbf{y}}_t$ is the **direction of $\tilde{\mathbf{H}}_\perp^t$** , which lies in the plane of incidence.

Then $\tilde{\mathbf{E}}_\perp^t$ is in the $\hat{\mathbf{y}}$ direction. The EM fields of the **transmitted wave** are

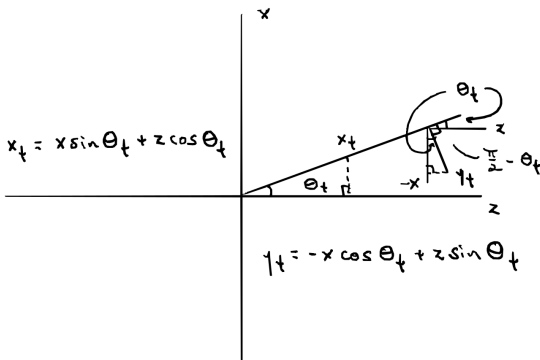
$$\begin{aligned}\tilde{\mathbf{E}}_\perp^t &= E_{\perp 0}^t e^{-jk_2 x_t} \hat{\mathbf{y}}, \\ \tilde{\mathbf{H}}_\perp^t &= \frac{E_{\perp 0}^t}{\eta_2} e^{-jk_2 x_t} \hat{\mathbf{y}}_t,\end{aligned}\quad (73)$$

where

$E_{\perp 0}^t = \tilde{\mathbf{E}}_\perp^t(x_t = 0)$ and k_2 and η_2 are the wavenumber and intrinsic impedance of Medium 2, respectively.



Reflection and Transmission at Oblique Incidence (Cont.)



Based on the geometry shown above,

$$x_t = x \sin \theta_t + z \cos \theta_t \quad \text{and} \quad \hat{y}_t = -\hat{x} \cos \theta_t + \hat{z} \sin \theta_t. \quad (74)$$

Reflection and Transmission at Oblique Incidence (Cont.)

So, in the **global** x - y - z **coordinate system**, the EM fields of the **transmitted wave** are

$$\tilde{\mathbf{E}}_{\perp}^t = E_{\perp 0}^t e^{-jk_2 x_t} \hat{\mathbf{y}} = E_{\perp 0}^t e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} \hat{\mathbf{y}} \quad (75)$$

and

$$\begin{aligned} \tilde{\mathbf{H}}_{\perp}^t &= \frac{E_{\perp 0}^t}{\eta_2} e^{-jk_2 x_t} \hat{\mathbf{y}}_t \\ &= \frac{E_{\perp 0}^t}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} (-\hat{\mathbf{x}} \cos \theta_t + \hat{\mathbf{z}} \sin \theta_t), \end{aligned} \quad (76)$$

where k_2 and η_2 are the wavenumber and intrinsic impedance of Medium 2, respectively.

Reflection and Transmission at Oblique Incidence (Cont.)

The **goal** is to use the above equations to **express** $\tilde{\mathbf{E}}_{\perp}^r$, $\tilde{\mathbf{H}}_{\perp}^r$, $\tilde{\mathbf{E}}_{\perp}^t$, and $\tilde{\mathbf{H}}_{\perp}^t$ **in terms of** $E_{\perp 0}^i$ **and** θ_i . This means finding relationships between the 4 quantities $E_{\perp 0}^r$, $E_{\perp 0}^t$, θ_r , and θ_t and the 2 parameters $E_{\perp 0}^i$ and θ_i .

What relationships connect these quantities, in general?
It's the **boundary conditions** at $z = 0$.

The **total electric field** in Medium 1 is

$$\tilde{\mathbf{E}}_{\perp}^1 = \tilde{\mathbf{E}}_{\perp}^i + \tilde{\mathbf{E}}_{\perp}^r \quad (77)$$

and the **total magnetic field** in Medium 1 is

$$\tilde{\mathbf{H}}_{\perp}^1 = \tilde{\mathbf{H}}_{\perp}^i + \tilde{\mathbf{H}}_{\perp}^r. \quad (78)$$

The boundary conditions state that the total electric and magnetic fields tangential to the boundary must be continuous across the boundary. Since the electric fields are all in the $\hat{\mathbf{y}}$ direction, this means

$$\left(\tilde{E}_{\perp y}^i + \tilde{E}_{\perp y}^r \right) \Big|_{z=0} = \tilde{E}_{\perp y}^t \Big|_{z=0}. \quad (79)$$

Reflection and Transmission at Oblique Incidence (Cont.)

Using the above expressions for the electric fields with $z = 0$, this implies

$$E_{\perp 0}^i e^{-jk_1 x \sin \theta_i} + E_{\perp 0}^r e^{-jk_1 x \sin \theta_r} = E_{\perp 0}^t e^{-jk_2 x \sin \theta_t}. \quad (80)$$

The magnetic fields only have components in the $\hat{\mathbf{x}}$ and $\hat{\mathbf{z}}$ directions and only the $\hat{\mathbf{x}}$ direction is tangential to the boundary. So, the tangential boundary condition for the magnetic field is

$$\left(\tilde{H}_{\perp x}^i + \tilde{H}_{\perp x}^r \right) \Big|_{z=0} = \tilde{H}_{\perp x}^t \Big|_{z=0}. \quad (81)$$

Setting $z = 0$ and using the above expressions for the magnetic fields, this means

$$-\frac{E_{\perp 0}^i}{\eta_1} \cos \theta_i e^{-jk_1 x \sin \theta_i} + \frac{E_{\perp 0}^r}{\eta_1} \cos \theta_r e^{-jk_1 x \sin \theta_r} = -\frac{E_{\perp 0}^t}{\eta_2} \cos \theta_t e^{-jk_2 x \sin \theta_t}. \quad (82)$$

Reflection and Transmission at Oblique Incidence (Cont.)

In summary, the boundary conditions imply

$$E_{\perp 0}^i e^{-jk_1 x \sin \theta_i} + E_{\perp 0}^r e^{-jk_1 x \sin \theta_r} = E_{\perp 0}^t e^{-jk_2 x \sin \theta_t}, \quad (83)$$

$$-\frac{E_{\perp 0}^i}{\eta_1} \cos \theta_i e^{-jk_1 x \sin \theta_i} + \frac{E_{\perp 0}^r}{\eta_1} \cos \theta_r e^{-jk_1 x \sin \theta_r} = -\frac{E_{\perp 0}^t}{\eta_2} \cos \theta_t e^{-jk_2 x \sin \theta_t}. \quad (84)$$

These equations must be true, regardless of the value of x . So, the exponents of the exponential functions must match. This implies the **phase-matching condition**

$$k_1 \sin \theta_i = k_1 \sin \theta_r = k_2 \sin \theta_t. \quad (85)$$

The first equality implies

$$\theta_r = \theta_i, \quad (\text{Snell's law of reflection}) \quad (86)$$

and the second equality implies

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{k_1}{k_2} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{n_1}{n_2}. \quad (\text{Snell's law of refraction}) \quad (87)$$

Reflection and Transmission at Oblique Incidence (Cont.)

Given the phase matching condition, the exponential terms in the boundary conditions are the same and can be divided out and, since $\theta_r = \theta_i$,

$$E_{\perp 0}^i + E_{\perp 0}^r = E_{\perp 0}^t, \quad (88)$$

$$\frac{\cos \theta_i}{\eta_1} (-E_{\perp 0}^i + E_{\perp 0}^r) = -\frac{\cos \theta_t}{\eta_2} E_{\perp 0}^t. \quad (89)$$

These equations can be solved for the **Fresnel reflection coefficient for perpendicular polarization**

$$\Gamma_{\perp} = \frac{E_{\perp 0}^r}{E_{\perp 0}^i} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \quad (90)$$

and the **Fresnel transmission coefficient for perpendicular polarization**

$$\tau_{\perp} = \frac{E_{\perp 0}^t}{E_{\perp 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}, \quad (91)$$

where θ_t is determined using Snell's law of refraction.

Reflection and Transmission at Oblique Incidence (Cont.)

As before, it's possible to show

$$\tau_{\perp} = 1 + \Gamma_{\perp}, \quad (92)$$

which is consistent with the balance of power across the boundary.

Useful special cases,

- When **Medium 2 is a perfect conductor**, $\eta_2 = 0$, $\Gamma_{\perp} = -1$, and $\tau_{\perp} = 0$. So, the wave is perfectly reflected.
- When the media are **nonmagnetic**, $\mu_1 = \mu_2 = \mu_0$ and

$$\Gamma_{\perp} = \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}. \quad (93)$$

Here, $\epsilon_2/\epsilon_1 = (n_2/n_1)^2$.

Homework

Read Chapter 8.

Exam 1 is on Thursday.

Homework 2 was due today.